EPSR-test-ex87 dockato: Qv site as first atom / first atom type

Program(s): dockato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

`dockato` [1] builds larger molecules from `.ato` building blocks: copies of the first molecule of a docking file are bonded onto free sites of a docked file, removing one atom of each to form the new bond. With the Qv site as the first atom and first atom type, its removal shifts the remaining atom types (shift branch of `clean_atom_types`).

Code exercised

Fortran (original, fortran_src/):

- `dockato.f`: `dockato` (driver, parameters from command line or stdin), `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`
- `dockato.f`: `test_overlap`, `bond_molecule` (ROT/DIH groups), `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`
- `ato_manipulation_rev4.f`: `get_direct_bonds`, `get_rotgrp`, `centreofmass`, `ftype`; `ato_read_and_save.f`: `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/dockato.rs`: `run`, `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`, `test_overlap`
- `tools/dockato.rs`: `bond_molecule`, `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`, `get_direct_bonds`, `get_rotgrp`
- `ato.rs`: `State::moloutc` (ROT/DIH groups), `State::ftype`; `rng.rs`: `Ran1::ran1`

Physical system

A water molecule (SPC/E parameters [2]) with a dummy `Qv` atom bonded to O1 at 1.45 Å, listed as atom 1. Atom `atomd` of the new molecule is placed on `atomv`, the molecule is rotated so that its `atomdr` neighbour lies on the `atomvr` neighbour of `atomv` (equal bond lengths required), an optional second (dihedral) rotation about the new bond uses `atomdrr`/`atomvrr` (random angle in [dminrr, dmaxrr] from `ran1`, the Park-Miller generator with Bays-Durham shuffle [3,4], or a fixed dihedral that creates a DIH group when dminrr = dmaxrr), overlaps are tested with a distance factor, `atomd` and `atomvr` are removed and the new rotational group, bonds and atom types are updated. As in ex86 the Qv atoms are removed and an O-O bond is formed.

Inputs

`WaterQvA.ato` created like `WaterQv.ato` of ex86 but with Qv as atom 1 and first atom type. Run 1: random 20-340°, factor 1.0, ndock 3 (only one site available); run 2: rotation only on the docking side (atomvrr = 0), overlaps allowed.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Each run writes one new `.ato` file (coordinates, bonds, rotational groups, atom types and the random-number state), compared byte for byte with the stored Fortran output. Both outputs hold 6 atoms (two O1, four H1, renumbered as types 1 and 2 after the Qv type is removed) and one ROT group.

Notes

Fortran behaviour reproduced: in `peroxideA.ato` the re-created Qv type gets the stale parameters of the last type slot (those of H1: mass 2, charge 0.4238).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).

[3] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).

[4] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).

[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.